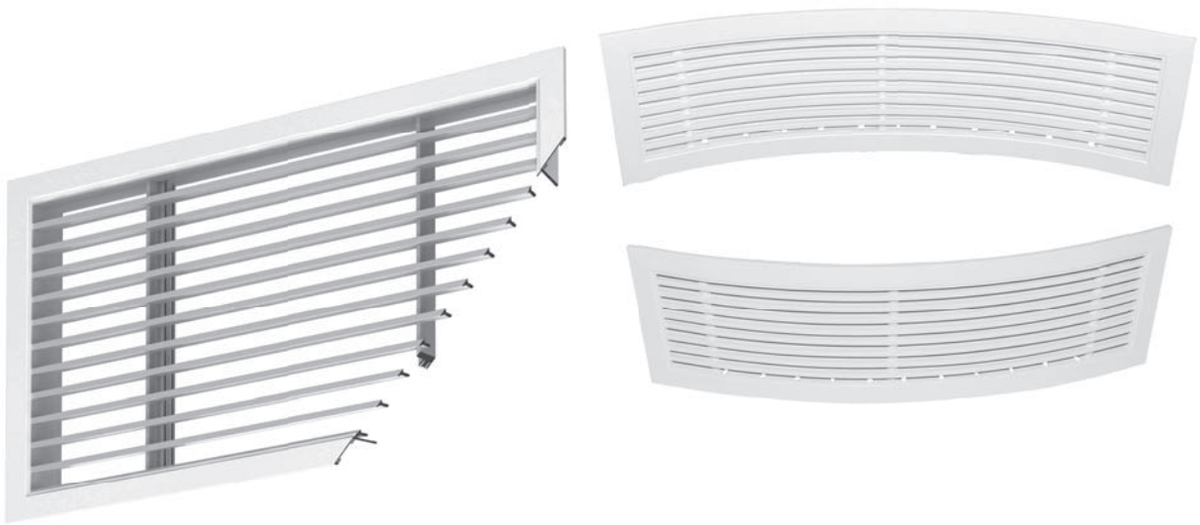
Linear Decorative Grille LBD



Description
Aluminum grille for supply/exhaust of the air flow, with 0 or 15 degree fixed blades.

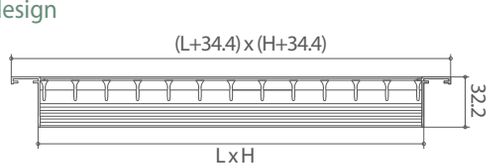
Application
Supply/exhaust of the air flow.

Characteristics
0 or 15 degree fixed blades, parallel with length "L".
The grille is provided with mounting holes (standard)
The air flow can be adjusted by mounting an air control damper OBD on the inlet connection.
On demand, the grille can be curved (concave/convex)

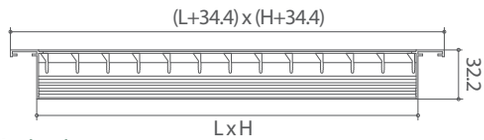
Accessories
Settlement frame made of galvanized sheet metal **CC**
Mounting spring **A.**
Air control damper with opposable blades **OBD**
Air filter **FIL**
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

Technical design
LBD - 0°



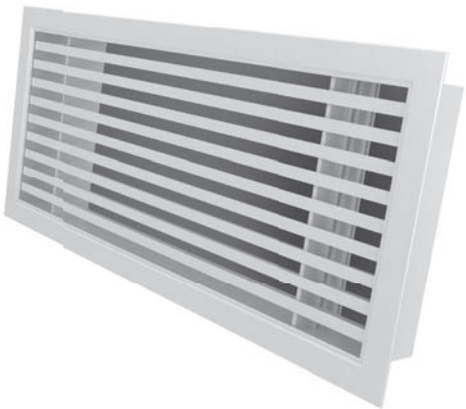
LBD - 15°



Assembly Methods
Screws (standard).
Settlement frame (optional)
Mounting spring (optional).

Linear Decorative Grille LBDB/LBDD





Description
Aluminum grille for supply/exhaust of the air flow, with 0 or 15 degree fixed blades.

Application
Supply/exhaust of the air flow.

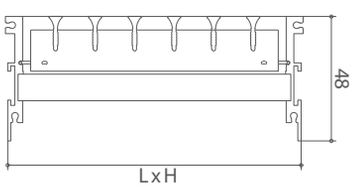
Characteristics
LBDB
- 0 or 15 degree fixed blades, parallel with length "L".
- Removable blades system
- Encased mounting into wall/ ceiling

LBDD
- Two rows of blades
- Frontal fixed blades at 0°/15°, parallel with lenght "L"
- Individually adjustable blades that allow air flow direction
- Provided with mounting holes (standard).

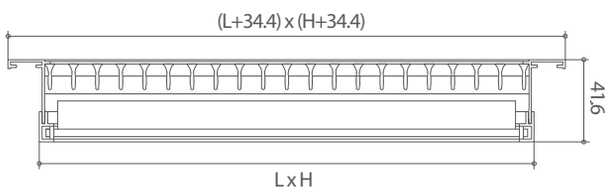
Accessories
Air control damper with opposable blades **OBD (LBDB)**
Mounting spring **A**
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design
LBDB



LBDD



Assembly Methods
LBDB - With screws on the inlet connection of the grille
LBDD - With screws on the frame (standard).
Mounting spring

Selection table LBD / LBDB / LBDD

Air flow m³/h	L x H	250x100	300x100	400x100	500x100	600x100	800x100	1000x100	1500x100	400x150	600x150	1000x150	600x200	800x200	1000x200
	Ak(m²)	0.011	0.013	0.019	0.024	0.029	0.041	0.05	0.075	0.033	0.045	0.081	0.064	0.085	0.12
140	X(m) NR (dB) ΔPt [pa] Veff (m/s)	3.0 <20 6.0 3.2													
200	X(m) NR (dB) ΔPt [pa] Veff (m/s)	4.2 25.0 10.0 4.6	3.8 21.0 10.0 4.0	3.2 <20 5.0 2.8											
300	X(m) NR (dB) ΔPt [pa] Veff (m/s)	6.3 35.0 30.0 6.9	5.8 30.0 18.0 6.0	5.0 24.0 10.0 4.2	4.3 18.0 7.0 3.3										
400	X(m) NR (dB) ΔPt [pa] Veff (m/s)	8.5 43.0 50.0 9.3	7.8 38.0 35.0 7.9	6.5 32.0 18.0 5.6	5.8 25.0 10.0 4.4	5.3 22.0 9.0 3.8	4.6 <20 5 2.6			5.1 25 8 3.4					
600	X(m) NR (dB) ΔPt [pa] Veff (m/s)	12.0 53.0 118.0 13.9	11.5 48.0 80.0 11.9	10.0 42.0 40.0 8.3	8.7 35.0 30.0 6.7	6.0 33.0 20.0 5.7	7 26 12 4.0	6 25 7 3.1		8 35 16 5.1	6.3 25 8 3.7				
800	X(m) NR (dB) ΔPt [pa] Veff (m/s)			15.0 51.0 75.0 11.1	12.0 45.0 48.0 8.9	10.5 45.0 35.0 7.7	9.3 33 20 5.3	8.5 32 15 4.2	6.5 20 6 3.2	11 42 32 6.7	8.5 30 15 4.9	6.8 21 5 2.7	7 25 8 3.5		
1000	X(m) NR (dB) ΔPt [pa] Veff (m/s)					13.0 45.0 55.0 9.6	11 42 30 6.6	10.2 37 20 5.2	8.3 28 9 4.0	13 46 48 8.4	11 38 22 6.2	8 27 8 3.3	9 30 12 4.3	7.8 25 8 3.3	7 <20 3 2.3
1400	X(m) NR (dB) ΔPt [pa] Veff (m/s)						16.5 50 55 9.3	14 45 37 7.3	12 37 18 5.6		15 48 40 8.6	12 35 15 4.7	12 40 22 6.1	11 33 12 4.6	11 33 12 3.2
2000	X(m) NR (dB) ΔPt [pa] Veff (m/s)											16 45 30 6.7	18 50 42 8.7	15 43 25 6.5	14 35 17 4.6
2500	X(m) NR (dB) ΔPt [pa] Veff (m/s)													20 48 37 8.2	18 45 25 5.8
3000	X(m) NR (dB) ΔPt [pa] Veff (m/s)														21 48 35 6.9

NR < 35 | 35≤NR≤45 | NR>45

Annotation

* Ak [m²] - Free surface
* X [m] - Air jet length
* Vk = 0.2 [m/s] - Velocity at which the air jet length was calculated "x" [m]
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss
* Veff[m/s] - Air velocity inside the grille

Order codes

Type - Blade's angle	+	CC	+	A (Mounting spring)	+	FIL	+	OBD	+	ADP IZ/N	/	Dimensions	/	O
LBD/LBDB / LBDD -0°/15°		Settlement frame (LBD)		No. of springs depending on grille dimensions		Air filter (LBD)		Air control damper (LBD/LBDB)		Insulated/ Non-insulated adapter		On request		**O** Painted white RAL 9016 **R** Other RAL paints Natural anodised (LBD)